

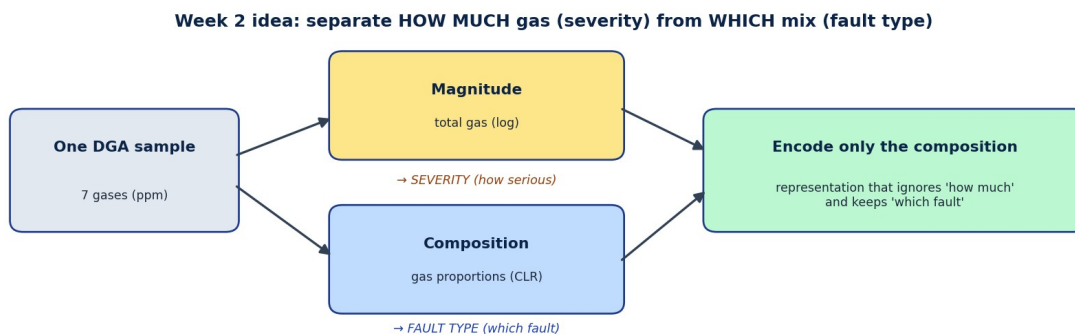
Week 2 Synthesis — Representation Learning (teaching the model what to look at)

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26 June 2026

1. This week in one paragraph

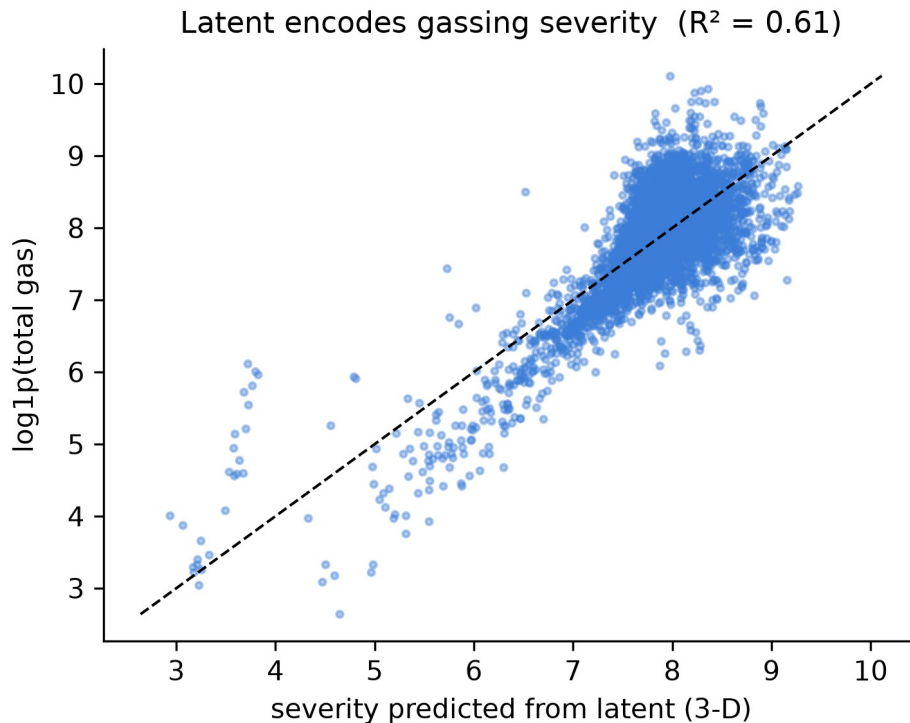
Week 2 was about **representation learning** — teaching the model to summarise each oil sample into a few numbers, and, crucially, to make those numbers capture the **fault type** (which fault), not just the **severity** (how much gas). I trained and tuned the autoencoder, compared a variational autoencoder (VAE), and ran the key experiment of the project: separating “how much” from “the mix”.



Week 2 idea: split each sample into magnitude (severity) and composition (fault type), and encode only the composition.

2. The problem from Week 1

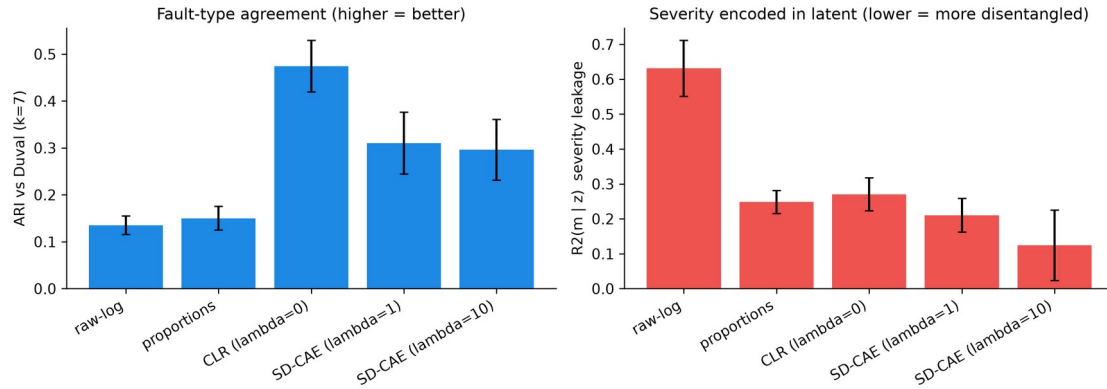
The plain autoencoder mostly learned **severity** — the overall amount of gas — not the fault type. The plot below shows it: the representation predicts total gas almost perfectly ($R^2 = 0.61$). That is useful for “how serious”, but it cannot tell partial discharge from thermal from arcing. The fix (already in the plan): fault type lives in the **proportions** of the gases — exactly what the Duval rule uses — so the model should look at proportions, not raw amounts.



The plain representation encodes severity: it predicts total gas almost perfectly ($R^2 = 0.61$), so it captures “how much”, not “which fault”.

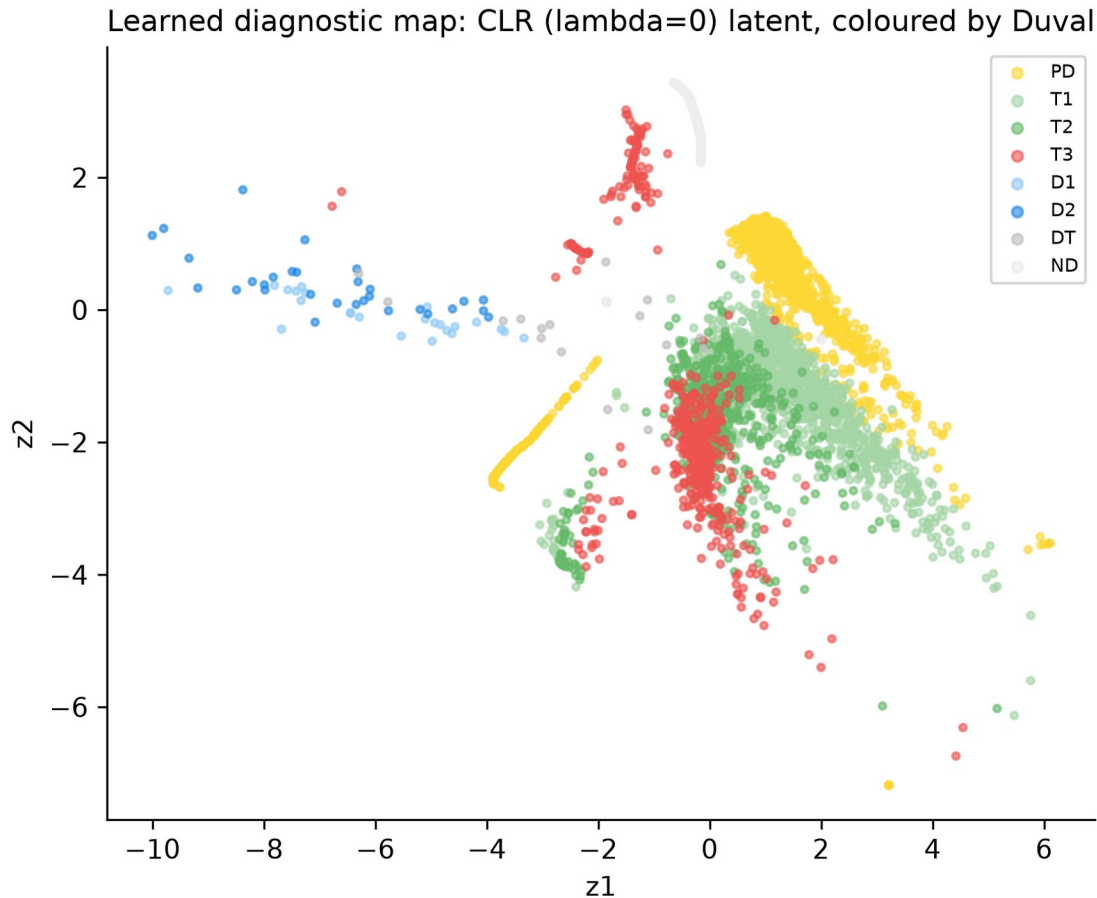
3. What I did (Week 2)

- **Trained and tuned the autoencoder** (different sizes and numbers of summary dimensions); checked training curves and reconstruction quality.
- **Implemented and compared a VAE** (a probabilistic autoencoder). The simpler compositional autoencoder gave the clearer, more interpretable separation, so I selected it.
- **Key experiment — separate magnitude from composition.** I described each sample by its gas **proportions** (a standard transform, CLR) and encoded only those — removing “how much” and keeping “which mix”.
- **Result:** agreement with the Duval fault types **more than tripled** — from **ARI 0.14** (raw amounts) to **0.47** (proportions), averaged over 5 runs — and severity stopped leaking into the representation (R^2 0.63 $\rightarrow$ 0.27).
- **Honest check (negative result):** I also tried an “adversarial” trick to force type and severity apart. It **did not help**, so the gain comes from the proportions idea itself, not the trick — reported as such.
- **Froze the chosen representation** and drafted the method section.



Agreement with Duval fault type (left, higher is better) more than triples with the compositional representation; severity leakage (right, lower is better) drops sharply.

The chosen representation now shows real **fault-type structure** — discharge (blue), thermal (green/red) and partial-discharge (yellow) samples fall in different regions:



The compositional representation, coloured by Duval type: discharge, thermal and partial-discharge samples separate into different regions.

On credibility: ARI measures **agreement with an expert rule (Duval)**, not accuracy against a true label — we have none. The honest claim is the **large relative gain (x3.4)** and the removal of severity leakage, not a perfect score. Rare fault classes (D1/D2) stay hard, as expected.

4. What I plan to do (Week 3)

Following the plan: take this representation into **clustering** (group the fleet into fault families) and **anomaly detection** (flag unusual samples), and **evaluate both quantitatively** — internal cluster quality, agreement with Duval, and checking flagged anomalies against the real field-event notes.

*All numbers are produced by code and saved under results/
(results/tables/sdcae_ablation.csv); nothing is hand-typed. Week-2 status — the
representation is frozen; clustering and anomaly evaluation come next.*